MODULE 6.3

Practical investigations

2 • Precipitation reactions

Planning & Conducting | Processing & Analysing

Purpose

To predict and then test particular precipitation reactions.

Timing

30 minutes

Materials

- 0.1 M solutions of silver nitrate, sodium carbonate, sodium hydroxide, barium nitrate and copper sulfate (all in 'dropper' bottles)
- 10 small test-tubes and test-tube rack
- gloves

SAFETY

The chemicals used in this investigation are toxic so in all cases avoid contact with skin, eyes, nose and mouth. Silver nitrate causes stains so avoid contact with skin and clothing. Sodium hydroxide is toxic and corrosive. Copper(II) sulfate is toxic. Barium nitrate is toxic and an irritant. Solutions from this experiment must not be washed down the sink. They should be placed in a clearly marked waste bottle.

Wear gloves, safety glasses and protective clothing at all times.

Procedure

1. Construct a table like the one below. You need 11 rows to record 10 tests. Give your table a title.
2. In your table, enter all possible combinations of two of the test solutions.
3. Using Table 6.3.2 on page 250, predict what should happen in each tube and write your prediction in the table.
4. For your first pair of solutions, place 10 drops of solution A in a test-tube, and then add 10 drops of solution B to the same test-tube (as shown in Figure 6.3.14). Place the tube in a test-tube rack and allow it to rest.
5. Repeat step 4 for each pair of solutions.
6. Dispose of all chemicals as directed by your teacher.

Results

Record all observations in your results table.

Review

1. Use the solubility rules in Table 6.3.2 on page 250 to determine what has precipitated from each solution.
2. Write word equations to describe what is happening in each case where a reaction occurred.

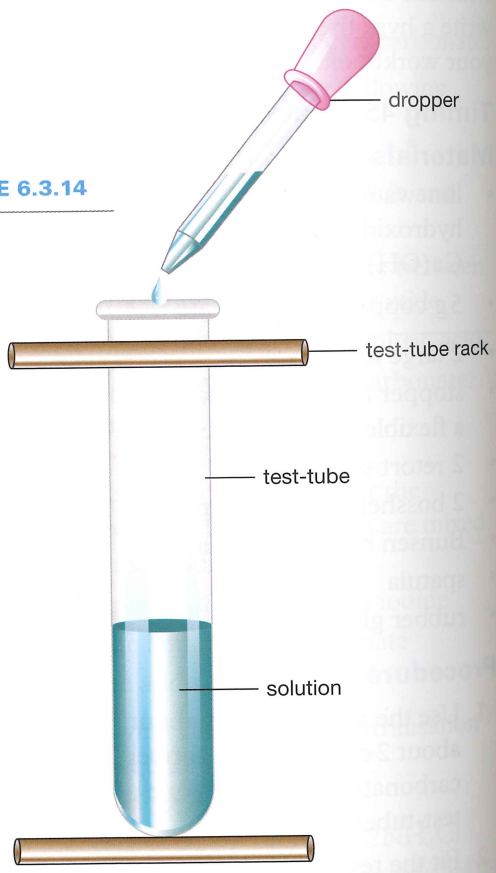


FIGURE 6.3.14

Solution A	Solution B	Prediction of precipitation (yes/no)	Observations before mixing	Observations after mixing	Name of precipitate (if any)
silver nitrate	sodium carbonate				